



RESEARCH ARTICLE

Integrated assessment of soil quality index, moisture index, fertility variability and erosion risk using geospatial techniques in the Guida-Kantabahal micro-watershed, Odisha

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Abstract

To ensure sustainable agricultural productivity in rainfed landscapes, a holistic understanding of soil quality, moisture availability, nutrient constraints and erosion vulnerability is required. In this study geospatial and geostatistical techniques were used to evaluation of soil fertility, soil quality index (SQI), soil moisture index (SMI) and erosion risk in the Guida-Kantabahal micro-watershed of Deogarh district, Odisha, was studied. The principal component analysis (PCA) based minimum dataset indicated that available sulphur (S), organic carbon (OC), available potassium (K), manganese (Mn) and available nitrogen (N) were the major determinants of soil quality which revealed that 84 % area in the watershed exhibited high soil quality (0.55–0.75), while the remaining 16 % displayed moderate soil quality (0.35–0.55). Across the watershed, 66.2 % of the area was moderately acidic, while the entire area was non-saline. A medium fertility status predominated covering 67.9 % of the area for OC, 56 % for N, 95 % for available phosphorus (P) and the entire watershed (100 %) for available K and 94.6 % low in sulphur (S) status. Micronutrient assessment revealed adequate availability of Fe, Mn, Cu and Zn across most of the watershed, whereas 67 % area was deficiency in B. The SMI was between 0.42 and 0.57, signifying moderate moisture availability. Revised universal soil loss equation (RUSLE)-based modelling estimated annual soil loss ranging from 0.2 to 510 t ha⁻¹ yr⁻¹ with an average value of 7.17 t ha⁻¹ yr⁻¹ while 10.36 % of the area experienced severe erosion (>100 t ha⁻¹ yr⁻¹). The integrated spatial assessment will help to rejuvenate the micro-watershed productivity in term of soil nutrient, moisture and soil conservation for sustainable land management.

Keywords: revised universal soil loss equation; soil moisture index; soil quality index; spatial variability; watershed

Introduction

Soil and water are vital natural resources that support agricultural production, environmental quality and ecosystem functioning. It regulates nutrient cycling, carbon storage and biological activity, thereby sustaining plant growth and ecosystem resilience under changing climatic conditions (1, 2). The current increasing pressure from intensive cultivation, inappropriate land-use practices, deforestation and climate variability has accelerated soil degradation, leading to declining productivity, reduced ecosystem services and growing threats to food security and sustainable land management (3). Among the various forms of land degradation, soil erosion, nutrient depletion, organic matter loss and moisture deficiency are major constraints affecting agricultural productivity worldwide (1, 3). These challenges are particularly severe in rainfed

regions, where crop production depends on the interaction of soil properties, topography and rainfall patterns (4). The loss of nutrient-rich topsoil due to soil erosion decreases the land's ability to store nutrients and water and affects soil functions, which results in a long-term loss of productivity of the land (1, 5). Moisture stress restricts nutrient availability and plant growth. Consequently, sustainable land management requires a comprehensive knowledge of the spatial distribution and interactions of these constraints within a watershed framework (6). Watersheds are the ideal spatial units for natural resource assessment because soil, water and land-use processes are highly integrated within their boundaries (7). Spatial variability in topography, parent material and land use generates large variations in soil properties and rates of degradation. Hence, watershed scale assessment through geospatial tools is crucial to detect problematic zones and advance the implementation of

sustainable land management (8). Soil quality is one of the most widely adopted concepts for assessing soil ability to maintain productivity and environmental functions in natural and managed ecosystems (9). Because none of these properties on their own can be used to represent overall soil functioning, quantitative soil health indices have been developed to combine multiple physical, chemical and biological indicators of soil health and sustainability into a single composite index (10). The use of Geographic Information System (GIS) and Global Positioning System (GPS) can help to accurately map the spatial variability of soil properties, providing a more detailed picture of the field-scale soil variability that would not be captured by traditional soil surveys. This spatial information can be used to develop site-specific management strategies, which helps to enhance the efficiency of resource-use and soil sustainability in long-term (11). Therefore, the soil quality index (SQI) and spatial variability assessment is one of the most popular indices used to assess soil health and sustainability. Recent studies have shown that SQI is a tool for assessing degraded areas, understanding management impacts and achieving sustainable land-use planning (11, 12). Multivariate statistical methods, like principal component analysis (PCA), have been of great benefit for the development of SQI, which enable the selection of minimal data sets by reducing the number of redundant soil variables (13–15) and identifying the most sensitive soil variables that control soil functions. Soil moisture also plays an important role in soil functions and crop productivity affecting nutrient availability, soil microbial activity, root growth and plant water uptake (16, 17). Soil quality assessment was conducted by (15) using PCA and geostatistical techniques; six key soil indicators were identified and ordinary kriging was used to map their spatial variability, revealing that more than 76 % of the watershed had high soil quality. In another study, previous researchers reported that 13.5 % (242.7 ha) of the study area had high soil quality, 25.7 % (462.8 ha) had medium soil quality and 53.8 % (966.1 ha) had low soil quality (14). The soil moisture index (SMI) provides a reliable measure of plant-available water and moisture stress, making it an important indicator for assessing drought vulnerability, particularly in rainfed regions (16–18). In addition to soil quality, fertility and moisture availability, soil erosion represents a critical yet under-reported factor affecting long-term land productivity and environmental sustainability (19). In this context geospatial techniques coupled with revised universal soil loss equation (RUSLE)-based models have been widely used to assess erosion susceptibility and identify priority areas for conservation planning. Among these, the RKLSCP factor, derived from the RUSLE, integrates the effects of rainfall erosivity (R), soil

erodibility (K), slope length and steepness (LS), cover-management (C) and support practice (P) to assess the combined influence of climate, soil properties, topography, land cover and conservation practices on potential soil erosion risk (20). This study integrates a physically based soil moisture index (SMI), a PCA-based SQI, geostatistical fertility mapping and RUSLE-based erosion assessment within a GIS framework to provide a comprehensive evaluation of watershed conditions. The novelty of this approach lies in deriving the SMI from field-measured soil moisture and laboratory -determined field capacity and permanent wilting point using a pressure plate apparatus, enabling site-specific assessment and sustainable management of soil, moisture and nutrient resources in the Guida-Kantabahal micro-watershed

Materials and Methods

Study area

The study was conducted in the year of 2023–24 at Guida-Kantabahal micro-watershed (725 ha of total geographical area) (Fig. 1) under the Reamal Sub-watershed (SWS Cluster), Deogarh District, Odisha, located at 21.3519° N latitude and 84.6610° E longitude. The climate of the area is tropical sub-humid with annual precipitation of 1453 mm, annual potential evapotranspiration (PET) of 1662.5 mm and length of growing period (LGP) is 160 days. Land use is dominated by scrubland (49.08 %), followed by agricultural land (33.22 %), forest (10.90 %), habitation (2.46 %), wasteland (1.25 %) and orchard plantations (1.10 %). The predominance of rainfed agriculture and diverse land-use patterns make the watershed suitable for evaluating soil quality, soil moisture status, fertility variability and susceptibility to erosion.

Soil sampling and analysis

The micro-watershed was mapped and digitised to a cadastral map for spatial analysis in GIS environment with a standard coordinate system. Soil samples were collected using a systematic grid design with a spacing of 320 × 320 m² following the semi-detailed Land Resource Inventory (1 : 8000 scale) developed by Indian Council of Agricultural Research–National Bureau of Soil Survey and Land Use Planning (ICAR–NBSS&LUP) (15, 19). A total of 70 geo-referenced composite (soils within 100 m were collected and mixed at those geo-points) soil surface samples (0–30 cm depth) were systematically collected to ensure adequate spatial representation across the watershed (10.24 ha/grid) (15). The collected samples were properly labelled, air-dried, ground and passed through a 2 mm sieve prior to laboratory analysis. The soil pH and electrical

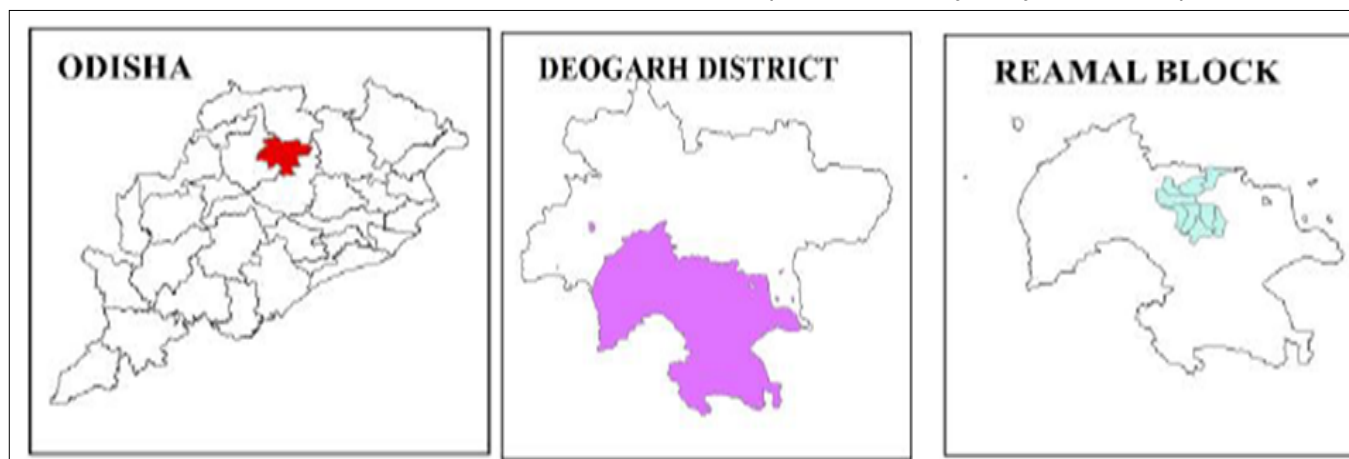


Fig. 1. Location map of the Guida-Kantabahal Micro-Watershed in Deogarh district, Odisha.

conductivity (EC) were determined in a 1 : 2.5 soil-to-water suspension while organic carbon (OC) was estimated by the Walkley and Black wet oxidation method (21). Available N was determined by the alkaline KMnO_4 method (22), while available P was estimated using Olsen's extraction method (23). Available K was measured using neutral normal ammonium acetate extraction followed by flame photometry (24). Available sulphur (S) was extracted with 0.15 % CaCl_2 solution (25) and determined turbidimetrically. Diethylenetriaminepentaacetic acid (DTPA)-extractable micronutrients, namely zinc (Zn), copper (Cu), iron (Fe) and Mn were analysed using an atomic absorption spectrophotometer (AAS) (26), whereas available boron (B) was determined using the hot water extraction method (27). Soil moisture content was measured *in situ* using a Time Domain Reflectometry (TDR) meter. Field capacity (FC) and permanent wilting point (PWP) were determined using a Pressure Plate Apparatus (28, 29).

Soil quality assessment

Soil quality index was developed using the minimum data set (MDS) approach through PCA in R-STUDIO (V2026.01.2-418). Principal components with eigenvalues ≥ 1 were retained and highly weighted variables with factor loadings within 10 % of the highest loading in each component were selected as MDS indicators (30). To avoid redundancy, significantly correlated variables were screened and only the most representative indicator was retained (31). The selected indicators were normalised to unitless scores (0–1) using appropriate linear scoring functions based on their contribution to soil quality ("more is better" or "less is better"). Indicator weights were derived from the proportion of variance explained by their respective principal components (15). Finally, the SQI was computed as the weighted sum of indicator scores (10, 14, 32):

$$\text{SQI} = \sum_{i=1}^n W_i S_i \quad \text{Eq. (1)}$$

where W_i is the weight assigned to the i th indicator based on the proportion of variance explained by its corresponding principal component and S_i is the normalised score of the i th soil quality indicator.

Soil moisture index

Soil moisture content was measured at each sampling location using a TDR meter (29) followed by FC and PWP in laboratory. The SMI was classified into five categories: very low (< 0.20), indicating severe moisture stress; low (0.20–0.40), indicating limited moisture availability; moderate (0.4–0.6), representing adequate moisture conditions for most crops; high (0.6–0.8), indicating good moisture availability; and very high (> 0.8), reflecting excessive to near-saturated soil moisture conditions. The SMI representing the relative availability of water to plants was calculated using the equation (17, 18):

$$\text{SMI} = \frac{\theta - \theta_{\text{PWP}}}{\theta_{\text{FC}} - \theta_{\text{PWP}}} \quad \text{Eq. (2)}$$

Where θ = measured volumetric soil moisture content (%)
 θ_{FC} = soil moisture content at Field Capacity (%) θ_{PWP} = soil moisture content at Permanent Wilting Point (%).

Estimation of soil loss

The RUSLE was integrated with GIS techniques used to estimate annual soil loss (20, 33, 34)). The RUSLE model predicts long-term average annual soil loss resulting from rainfall and associated overland flow and is expressed as:

$$A = R \times K \times LS \times C \times P \quad \text{Eq. (3)}$$

where A is the average annual soil loss ($\text{t ha}^{-1} \text{ yr}^{-1}$), R is the rainfall erosivity factor, K is the soil erodibility factor, LS is the slope length and slope steepness factor, C is the cover-management factor and P is the conservation practice factor. The R factor was estimated from long-term mean annual rainfall using the empirical equation proposed by Bergsma (35). The K factor was calculated following Wischmeier and Smith (33) using soil texture, organic matter, soil structure and permeability. The LS factor was derived from the Digital Elevation Model (DEM) based on slope and flow accumulation using the Moore and Burch (36) approach in ArcGIS. The C factor was estimated from the Normalized Difference Vegetation Index (NDVI) derived from satellite imagery using the equation proposed by previous researchers (37) whereas the P factor was assigned according to the prevailing land-use pattern and soil conservation practices. Then these raster layers were integrated in ArcGIS to estimate the spatial distribution of annual soil loss across the watershed, expressed as tonnes per hectare per year ($\text{t ha}^{-1} \text{ yr}^{-1}$).

Spatial variability mapping and cross validation

The spatial variability of SQI, SMI and soil nutrient parameters was analysed using Ordinary Kriging in ArcGIS 10.4. Experimental semivariograms were developed to quantify spatial dependence among observations. The best-fit semivariogram model was selected based on cross-validation statistics and interpolation performance was evaluated using mean error (ME), mean standardised error (MSE), root mean square error (RMSE), average standard error (ASE) and root mean square standardised error (RMSE) (38). The degree of spatial dependence was assessed using the nugget to sill ratio, where values < 0.25 , 0.25–0.75 and > 0.75 indicated strong, moderate and weak spatial dependence, respectively (39, 40). The selected model was subsequently used to generate continuous spatial distribution maps of the studied parameters across the watershed.

$$\text{Semi-variogram: } \gamma(h) = 1/(2N(h)) \sum [Z(x_i) - Z(x_i+h)]^2 \quad \text{Eq. (4)}$$

Where, $Z(x_i)$ = magnitude of the variable Z at location x_i , h = lag distance and $N(h)$ = number of paired sample points segregated by h .

$$\text{RMSE} = \sqrt{[(1/N) \sum (Z(x_i) - \hat{z}(x_i))^2]} \quad \text{Eq. (5)}$$

Where $Z(x_i)$ = observed value at location x_i $\hat{z}(x_i)$ = predicted value at location x_i N = total number of observations.

Statistical analysis and spatial modelling

Statistical analyses including correlation, descriptive statistics and PCA were performed using R-Studio (Version 2026.01.2-418; Posit Software, Boston, MA, USA) with statistical significance assessed at $p \leq 0.05$. Spatial interpolation and map generation were carried out in ArcGIS 10.4 (Esri Inc., Redlands, CA, USA) using ordinary kriging.

Results and Discussion

Principal component analysis and soil quality index development

The PCA identified five principal components (PCs) with eigenvalues greater than 1, collectively explaining 60.12 % of the total variance in the measured soil properties (Table 1) (Fig. 2 and 3). Based on the factor loading criteria, five indicators were selected for the MDS (Table 2): available S (PC1), OC (PC2), available K (PC3), Mn (PC4) and available N (PC5) (14). The descriptive statistics of the soil properties

Table 1. Principal components of soil quality parameters, eigenvalues and component matrix variables

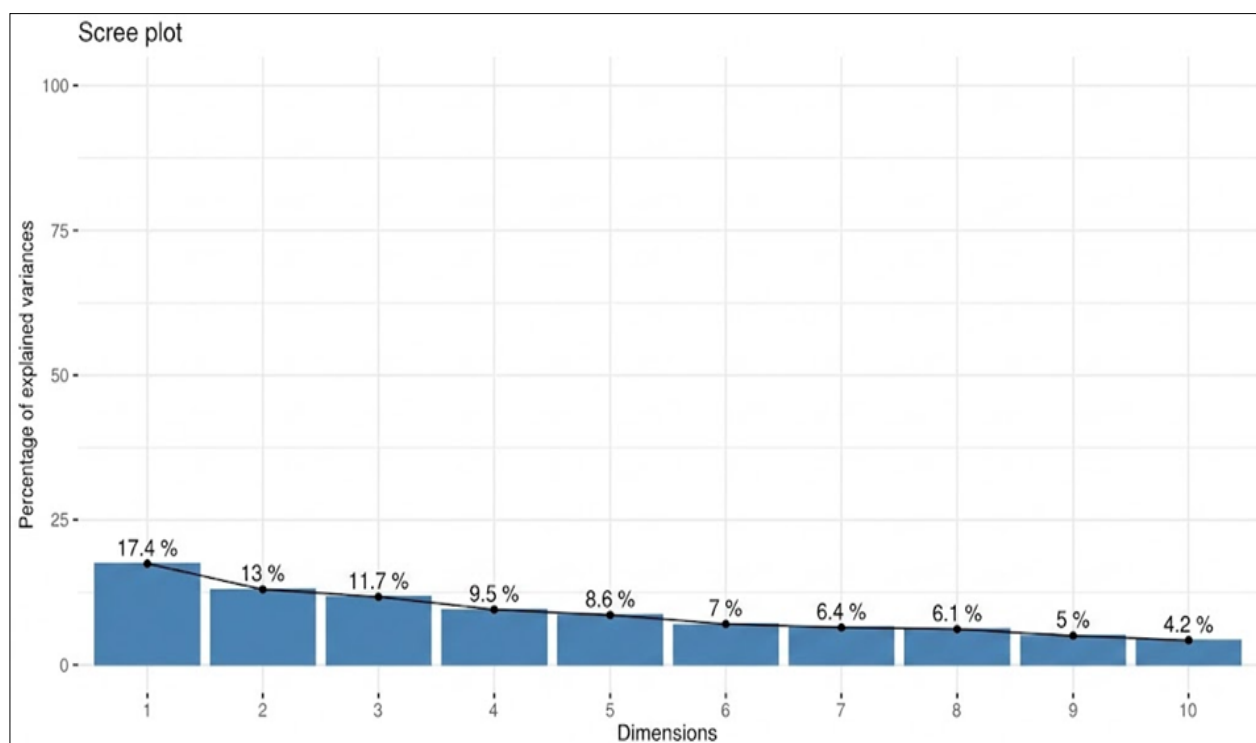
Principal components	PC1	PC2	PC3	PC4	PC5
Eigen values	2.440	1.814	1.637	1.326	1.198
% variance	17.432	12.957	11.696	9.469	8.561
Cumulative %	17.432	30.389	42.085	51.554	60.115
Weightage factor	0.290	0.216	0.195	0.157	0.142
Factor loadings (Rotated component matrix)					
pH	0.057	-0.296	0.075	-0.182	0.236
EC	-0.175	-0.267	0.489	-0.261	-0.057
OC	-0.264	-0.440	0.114	0.158	0.226
N	0.079	0.191	0.172	0.130	0.669
P	-0.401	0.234	0.258	0.046	-0.030
K	-0.042	0.123	-0.534	0.349	0.010
Ca	-0.252	-0.388	-0.130	0.365	-0.058
Mg	0.036	-0.277	-0.308	0.057	0.478
S	-0.484	-0.118	-0.054	0.014	0.021
Zn	-0.423	0.357	0.000	0.207	-0.027
Fe	-0.197	-0.100	-0.389	-0.406	-0.242
Mn	-0.144	-0.066	-0.302	-0.539	0.160
Cu	-0.193	0.398	-0.064	-0.312	0.350
B	-0.392	0.021	0.003	0.012	0.048

EC: electrical conductivity; OC: organic carbon; N: nitrogen; P: phosphorus; K: potassium; Ca: calcium; Mg: magnesium; S: sulphur; Zn: zinc; Fe: iron; Mn: manganese; Cu: copper; B: boron. *Bolded ones highly weighted variables in each PC, consider for minimum data set.

Table 2. Minimum dataset (MDS) variables selected through principal component analysis (PCA) for soil quality assessment

PC	Indicator	Loading	Weight
PC1	S	-0.484	0.290
PC2	OC	-0.440	0.216
PC3	K	-0.534	0.195
PC4	Mn	-0.539	0.157
PC5	N	0.669	0.142

PC: principal component; S: sulphur; OC: organic carbon; K: potassium; Mn: manganese; N: nitrogen.

**Fig. 2.** Scree plot of principal component analysis (PCA).

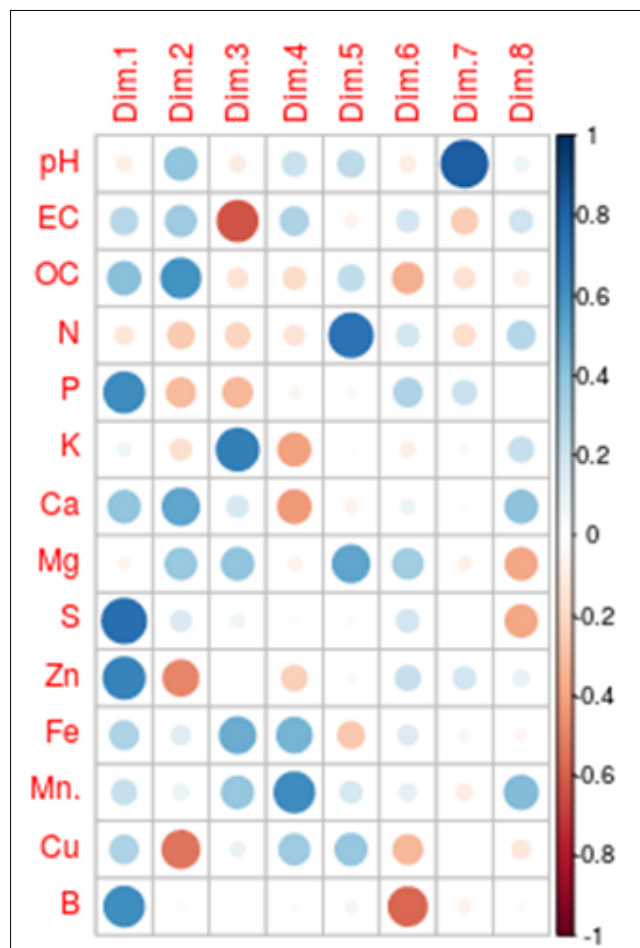


Fig. 3. Correlation matrix showing relationships among soil properties used in principal component analysis (PCA).

used in the PCA and soil quality assessment are presented in Table 3. The PCA-based SQI reduced data redundancy while preserving the most informative indicators for reliable soil quality assessment (14). Although pH, EC, available phosphorus (P), Zn, B, calcium (Ca) and

magnesium (Mg) are recognised as important soil quality indicators, they were not retained in the MDS because they exhibited relatively low factor loadings or showed strong correlations with the selected variables, indicating that they contributed redundant information (31). The contributions of the selected indicators to the SQI are presented in Table 4. The normalised scores were subsequently integrated through a weighted additive model to compute the SQI:

$$SQI = (0.290 \times S_s) + (0.216 \times OC_s) + (0.195 \times K_s) + (0.157 \times Mn_s) + (0.142 \times N_s) \quad \text{Eq. (6)}$$

where S_s , OC_s , K_s , Mn_s and N_s denote the normalised scores of available S, OC, available K, Mn and N, respectively. The OC was a dominant SQI indicator because it improves nutrient cycling, soil structure and water retention, thereby supporting crop productivity and climate resilience (9, 12), while S was retained due to its key role in protein synthesis and N use efficiency (10, 14). The PCA successfully reduced the number of indicators while retaining the major variability in soil properties reported by previous researchers (14). However, the selected MDS differed because soil quality indicators were influenced by local soil and land-use conditions.

The computed SQI values ranged from 0.468 to 0.832, with a mean of 0.626 (Table 5). The interpolated SQI map (Fig. 4) showed that 84 % of the watershed was under the high SQI class (0.55–0.75), while the remaining 16 % was classified as moderate (0.35–0.55). This indicates that the soil resource base is largely capable of sustaining productive agriculture, while the moderate-quality zones highlight areas needing targeted management (13, 15). The predominance of high soil quality in the present watershed agrees with previous studies, which also reported generally good SQI across an irrigated command area (13). Likewise, previous researchers demonstrated that PCA-based SQI effectively delineates spatial variations in soil quality for sustainable land management (14).

Table 3. Descriptive statistics of selected soil quality indicators used for principal component analysis (PCA) and soil quality

Parameter	Unit	Minimum	Maximum	Mean $\pm$ SE	Median	CV
pH	–	4.60	7.30	5.94 $\pm$ 0.05	5.90	7.6
EC	dS m ⁻¹	0.013	0.760	0.103 $\pm$ 0.018	0.054	147.6
OC	%	0.250	0.790	0.530 $\pm$ 0.019	0.545	29.4
Available N	kg ha ⁻¹	185.0	448.0	258.77 $\pm$ 7.13	233.00	23.1
Available P	kg ha ⁻¹	6.0	45.0	22.28 $\pm$ 1.58	13.85	59.4
Available K	kg ha ⁻¹	121.0	293.4	212.95 $\pm$ 5.23	213.00	20.5
Exchangeable Ca	cmol(+) kg ⁻¹	0.570	3.140	1.044 $\pm$ 0.044	1.042	35.0
Exchangeable Mg	cmol(+) kg ⁻¹	0.100	0.920	0.546 $\pm$ 0.025	0.525	37.8
Available S	kg ha ⁻¹	9.60	30.20	16.02 $\pm$ 0.58	14.90	30.1
Available Zn	mg kg ⁻¹	0.210	2.400	0.951 $\pm$ 0.063	0.775	55.6
Available Fe	mg kg ⁻¹	4.50	7.10	5.90 $\pm$ 0.08	6.10	11.9
Available Mn	mg kg ⁻¹	2.89	6.90	4.60 $\pm$ 0.12	4.23	21.0
Available Cu	mg kg ⁻¹	3.44	8.80	6.11 $\pm$ 0.12	5.98	15.9
Available B	mg kg ⁻¹	0.120	0.920	0.459 $\pm$ 0.022	0.410	40.9

EC: electrical conductivity; OC: organic carbon; N: nitrogen; P: phosphorus; K: potassium; S: sulphur; Zn: zinc; Fe: iron; Mn: manganese; Cu: copper; B: boron; SE: standard error; CV: coefficient of variation.

Table 4. Contributions of significant soil parameters to soil quality index

Parameters	Contribution to SQI	Minimum	Maximum	Mean	SD	CV (%)
Available S	0.290 $\times$ more is better	9.6	30.2	16	4.82	30.1
OC	0.215 $\times$ more is better	0.25	0.79	0.53	0.15	29.4
Available K ₂ O	0.195 $\times$ more is better	121	293.4	212.9	43.74	20.5
Mn	0.157 $\times$ optimum is better	2.9	6.9	4.6	0.96	21
Available N	0.142 $\times$ more is better	185	448	258.8	59.6	23.1
SQI	Weighted additive index	0.468	0.832	0.626	0.083	13.18

SQI: soil quality index; OC: organic carbon; K: potassium; Mn: manganese; N: nitrogen; S: sulphur; SD: standard deviation; CV: coefficient of variation.

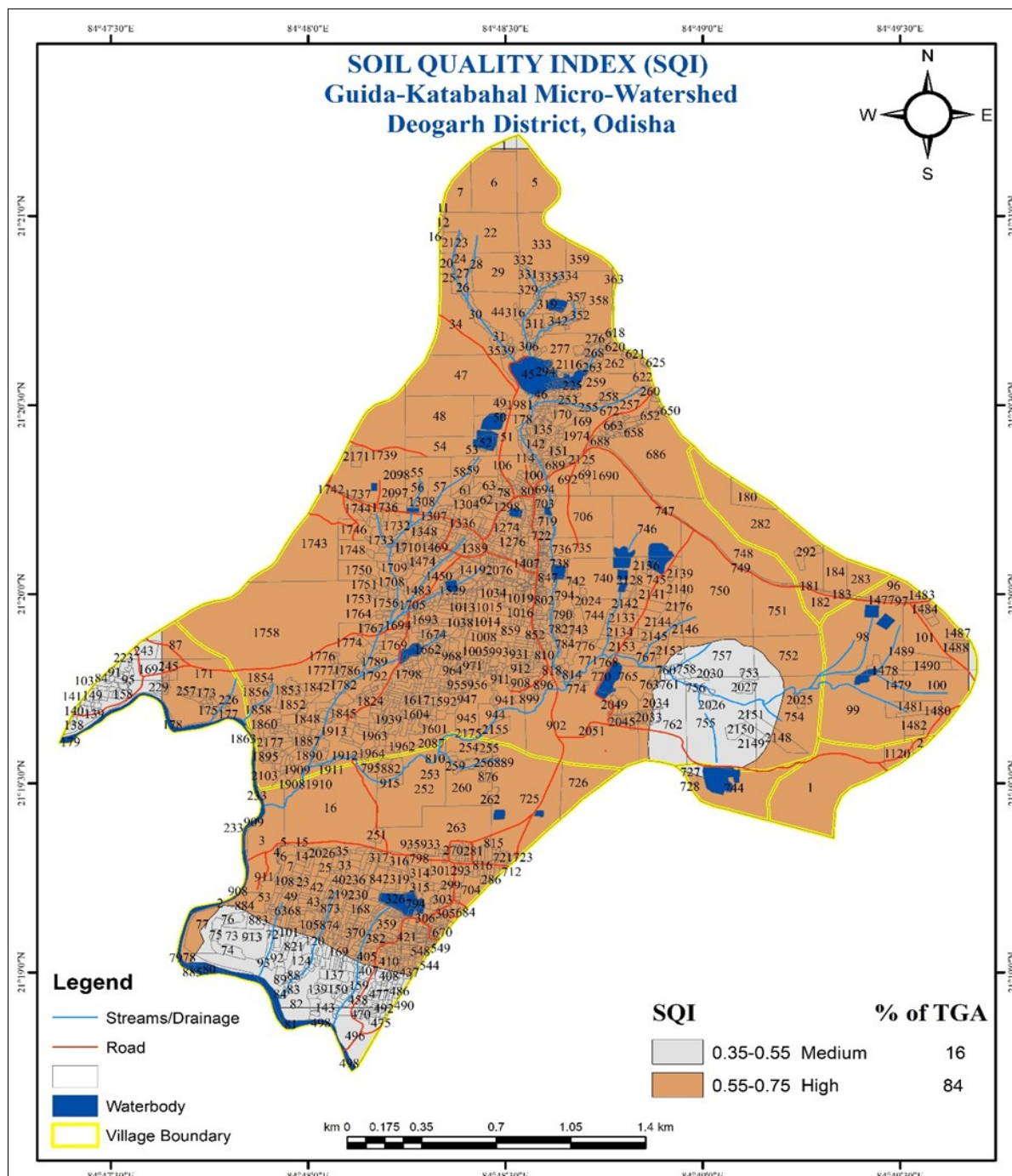


Fig. 4. Spatial distribution of soil quality index (SQI) across the Guida-Kantabahal Micro-Watershed.

Soil moisture index

The SMI (Fig. 5) ranged from 0.42 to 0.57, with a mean value of 0.496 (Table 5), indicating moderate soil moisture availability across the watershed. The moderate moisture status indicates that soils held almost half of available water between field capacity and permanent wilting point, which is suitable for crop growth (17). Soil texture, OC, topography and water-holding capacity (18) were the most important factors affecting variations in SMI. Finer textured, organic matter-rich soils had higher SMI values and lower values were correlated with a relatively lower moisture retention (29). The moisture regime observed is adequate for crop production, but *in situ* moisture conservation practices such as mulching, contour bunding, farm bunding and farm ponds should be adopted to improve soil moisture availability during dry periods (18, 29).

Annual soil loss estimation

The produced soil loss map showed significant spatial variation in erosion over the watershed (Fig. 6). The estimated annual soil loss ranged from 0.2 to 510 t ha⁻¹ yr⁻¹, with a mean value of 7.17 t ha⁻¹ yr⁻¹ excluding high-rate outliers from localised macro-basin sectors. The spatial distribution of soil loss indicated that 43.5% of the watershed area fell within the 10–25 t ha⁻¹ yr⁻¹, followed by 26.86 % of the watershed falls under 25–100 t ha⁻¹ yr⁻¹. The watershed had 16.50 % of moderate (5–10 t ha⁻¹ yr⁻¹) erosion areas and 10.36 % of severe erosion (> 100 t ha⁻¹ yr⁻¹) areas. The watershed was characterised as low erosion (< 5 t ha⁻¹ yr⁻¹) area, accounting for only 2.77 % of the watershed. The area was dominated by low to moderate soil erosion classes, corresponding to relatively stable land conditions, with extreme soil erosion hotspots (>100 t ha⁻¹ yr⁻¹) being associated with

Table 5. Soil moisture index (SMI)

Parameter	Minimum	Maximum	Mean	SD	CV (%)
SMI	0.28	0.81	0.496	0.117	23.59

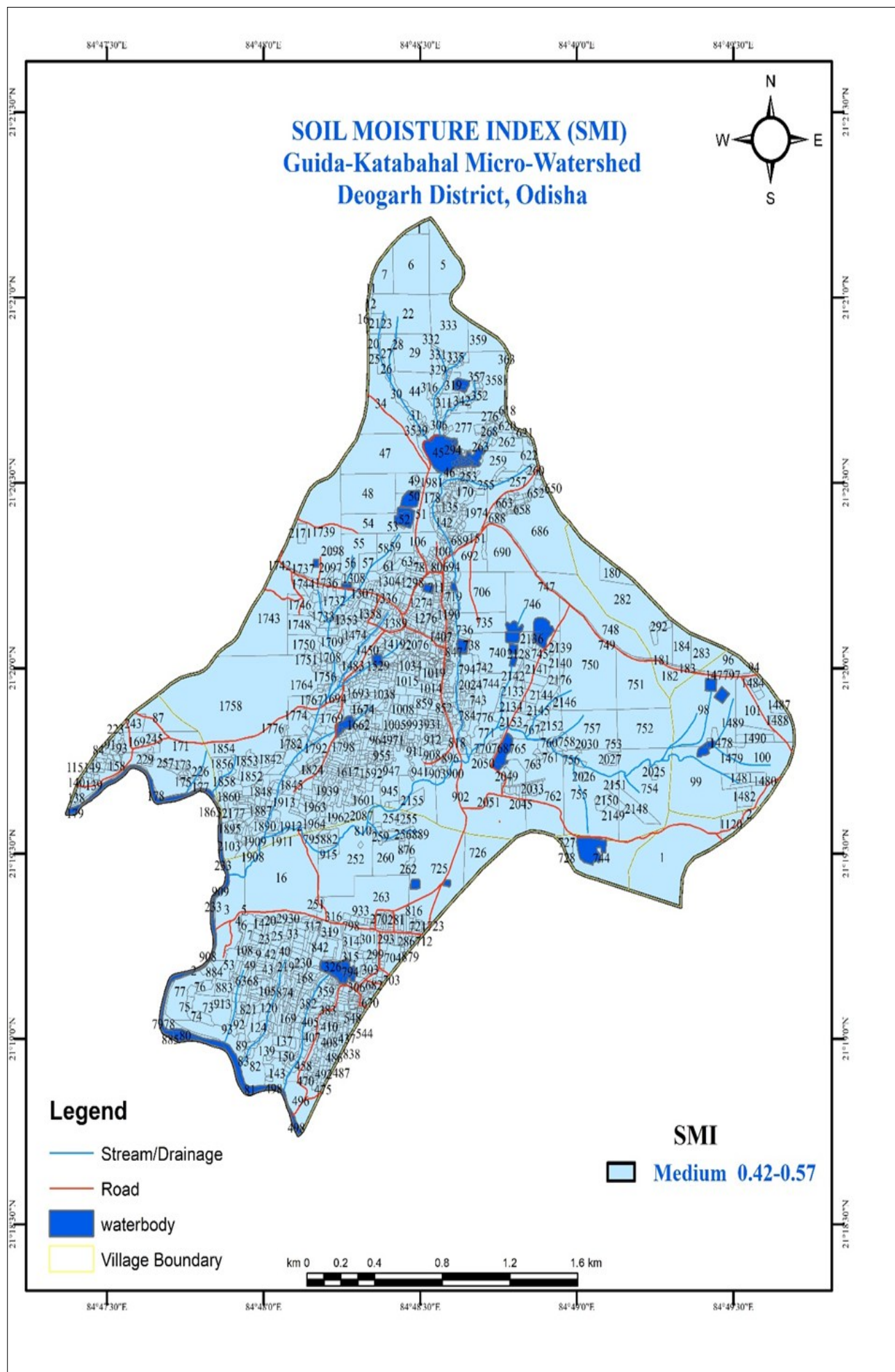


Fig. 5. Spatial distribution of soil moisture index (SMI) across the Guida-Katabahal micro-watershed.

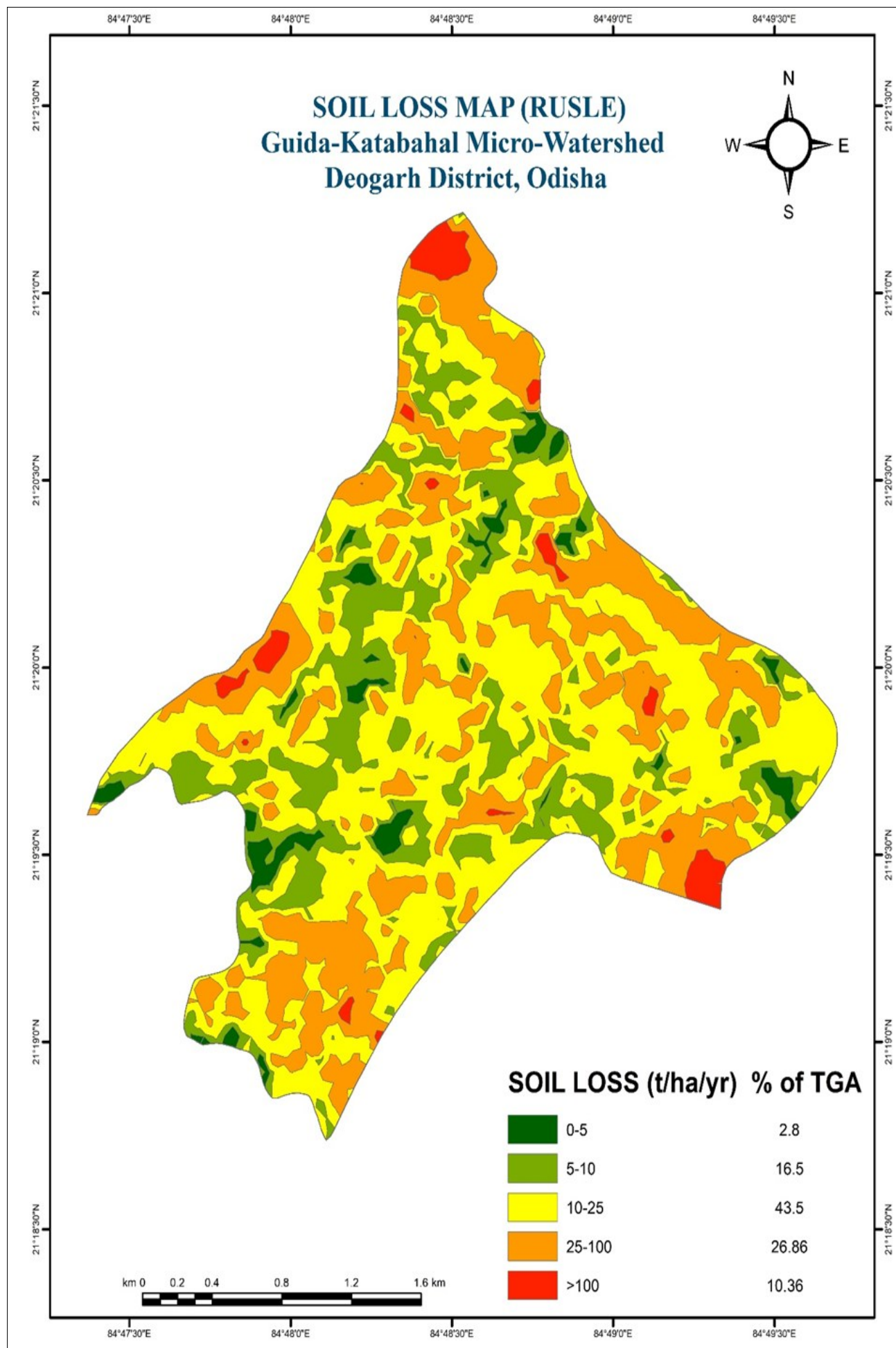


Fig. 6. Spatial distribution of estimated annual soil loss based on the revised universal soil loss equation (RUSLE) across the Guida-Kantabahal micro-watershed.

steep slopes, low vegetation cover and concentrated runoff areas (33). Similar trends have been reported in other RUSLE-based watershed studies, where topography, land use and rainfall erosivity were identified as the primary drivers of soil erosion (33, 34). The occurrence of severe erosion over 10.36 % of the watershed indicates that contour cultivation, vegetative barriers and cover crops should be prioritised in these areas to reduce soil loss and prevent further land degradation (19). Although RUSLE is a widely accepted tool for erosion assessment, the predicted soil loss should be interpreted as an estimate of potential erosion because field-based sediment yield or erosion plot data were not available for validation. Despite this limitation, the model provides a reliable basis for identifying erosion-prone areas and prioritising conservation planning (19, 20).

Spatial distribution maps of soil quality index, soil moisture index and soil loss

For SQI (Fig. 4), the spherical semivariogram model exhibited excellent predictive performance with a regression equation of $y = 0.9996x + 0.0003$, $ME = 0.00002$, $RMSE = 0.00118$, $ASE = 0.00423$ and $MSE = 0.0058$. The MSE was close to zero, indicating negligible prediction bias. The nugget ratio of 0.23 suggested strong spatial dependence, indicating that the observed variability was predominantly controlled by intrinsic soil-forming factors rather than random variation (13). Similarly, the SMI (Fig. 5) interpolation showed high predictive accuracy with a regression equation of $y = 0.9968x + 0.0016$, $ME = 0.000002$, $RMSE = 0.00275$, $ASE = 0.00352$ and $MSE = -0.0011$. The RMSSE (0.955) was close to the ideal value of one, confirming the robustness of the interpolation model. The nugget ratio of 0.18 further indicated strong spatial continuity of soil moisture conditions within the watershed. The soil loss map (Fig. 6) was also best described by the spherical semivariogram model. Cross-validation yielded a regression equation of $y = 0.9322x + 4.1885$, $ME = 2.94$, $RMSE = 14.72 \text{ t ha}^{-1} \text{ yr}^{-1}$, $ASE = 37.06$ and $MSE = 0.171$, indicating acceptable predictive performance despite the greater spatial variability associated with erosion processes. The nugget ratio of 0.31 suggested moderate spatial dependence, implying that both intrinsic factors (soil properties, topography and land cover) and extrinsic factors (rainfall variability and land management practices) influenced soil erosion patterns (38, 39).

Spatial distribution of soil fertility attributes

Geostatistical analysis revealed distinct spatial dependence among the studied soil fertility parameters. The Gaussian model provided the best fit for N, while the Exponential model was most suitable for available P and OC. The Spherical model effectively figures adequately captured the spatial dependence of soil properties and enabled reliable fertility mapping through ordinary kriging (39). The soil pH map revealed that the watershed was predominantly acidic in nature (Fig. 7). Moderately acidic soils pH (5.6–6.0) occupied the largest proportion of the area (66.2 %), followed by strongly acidic soils (5.0–5.5) covering 19.4 %. Slightly acidic soils (6.1–6.5) accounted for 12.0 %, while strongly acidic (4.5–5.0) and neutral soils (6.6–7.0) occupied only 0.8 % and 1.6 % of the watershed area, respectively. These results indicate that soil acidity is a dominant constraint affecting nutrient availability and crop productivity in the watershed (2, 40). The EC values remained below 0.2 dS m^{-1} throughout the watershed, indicating that the micro watershed was non-saline and free from salinity-related limitations (Fig. 8). Conversely the soil OC status was predominantly medium in 67.9 % of the area and the remaining 32.1 % exhibited low OC content (Fig. 9). Available N (Fig.

10) with 56 % of the watershed falling under the medium category ($250\text{--}500 \text{ kg ha}^{-1}$) and 44 % under the low category ($< 250 \text{ kg ha}^{-1}$). The absence of high N status indicates the need for balanced nutrient management to sustain crop productivity (12, 41). Available P was generally adequate across the watershed, with 95.7 % of the area classified under the medium category ($14\text{--}40 \text{ kg ha}^{-1}$) and only 4.3 % under the low category ($< 14 \text{ kg ha}^{-1}$) suggests that P availability is unlikely to be a major limiting factor in most parts of the watershed (41) (Fig. 11). Available K (Fig. 12) was uniformly distributed within the medium fertility class ($118\text{--}280 \text{ kg ha}^{-1}$), covering the total watershed area indicates relatively stable potassium reserves capable of supporting crop growth under current management practices (41, 42). The low S availability (94.6 % of TGA) emerged as a major nutritional constraint for crop production (Fig. 13). Among the micronutrients, most of the watershed was adequately supplied with Fe, Mn, Cu and Zn (Fig. 14–17), however, B (Fig. 18) deficiency was prominent, affecting 67 % of the area (42). The observed spatial variability in soil fertility reflects the combined influence of soil acidity, parent material, rainfall and long-term land use practices on nutrient distribution within the watershed (43). Predominantly acidic soils may reduce nutrient availability and fertiliser-use efficiency, highlighting the need for site-specific liming together with balanced fertilisation in strongly acidic areas to improve soil reaction (41, 42). Medium OC and N status suggest emphasising the importance of farmyard manure, crop residue incorporation and green manuring to enhance soil OC in low to medium OC areas (6, 12). The adequate availability of P and K indicates that these nutrients can be managed through soil test-based fertiliser recommendations rather than blanket fertiliser applications (41, 43). In contrast, the widespread deficiencies of S and B are likely associated with continuous crop removal, acidic conditions, high rainfall and highly weathered parent materials that promote nutrient leaching (25, 27). Therefore, targeted application of S and B containing fertilisers, integrated with organic amendments, should be prioritised identified deficient areas. The combined interpretation of the SQI (Fig. 4), SMI (Fig. 5), soil erosion (Fig. 6) and soil fertility maps (Fig. 7–18) provides a decision-support framework for watershed management. High-quality soils with adequate moisture and low erosion risk can be prioritised for intensive cultivation, whereas nutrient-deficient acidic areas require liming and balanced fertilisation and erosion-prone locations should be managed through soil and water conservation practices.

Conclusion

The integrated modelling of soil quality, water availability, fertility status and erosion risk within a geospatial framework provided a comprehensive evaluation of land resource potential in the Guida-Kantabahal micro watershed. High soil quality index (SQI) values and moderate soil moisture index (SMI) conditions suggest an overall favourable soil environment for plant growth, but a large spatial variability of soil properties suggests a need for management efforts specific to location. The main constraints on sustainable productivity were identified as moderately acidic soil reaction, moderate to severe boron (B) deficiency, low availability of sulphur (S) and severe erosion in certain areas. The study also shows the usefulness of the use of minimum datasets (MDS) derived from principal component analysis (PCA), soil quality assessment, geo-statistics and erosion modelling to characterise spatial variability and identify critical resource constraints at the watershed scale. This integrated approach enabled the delineation

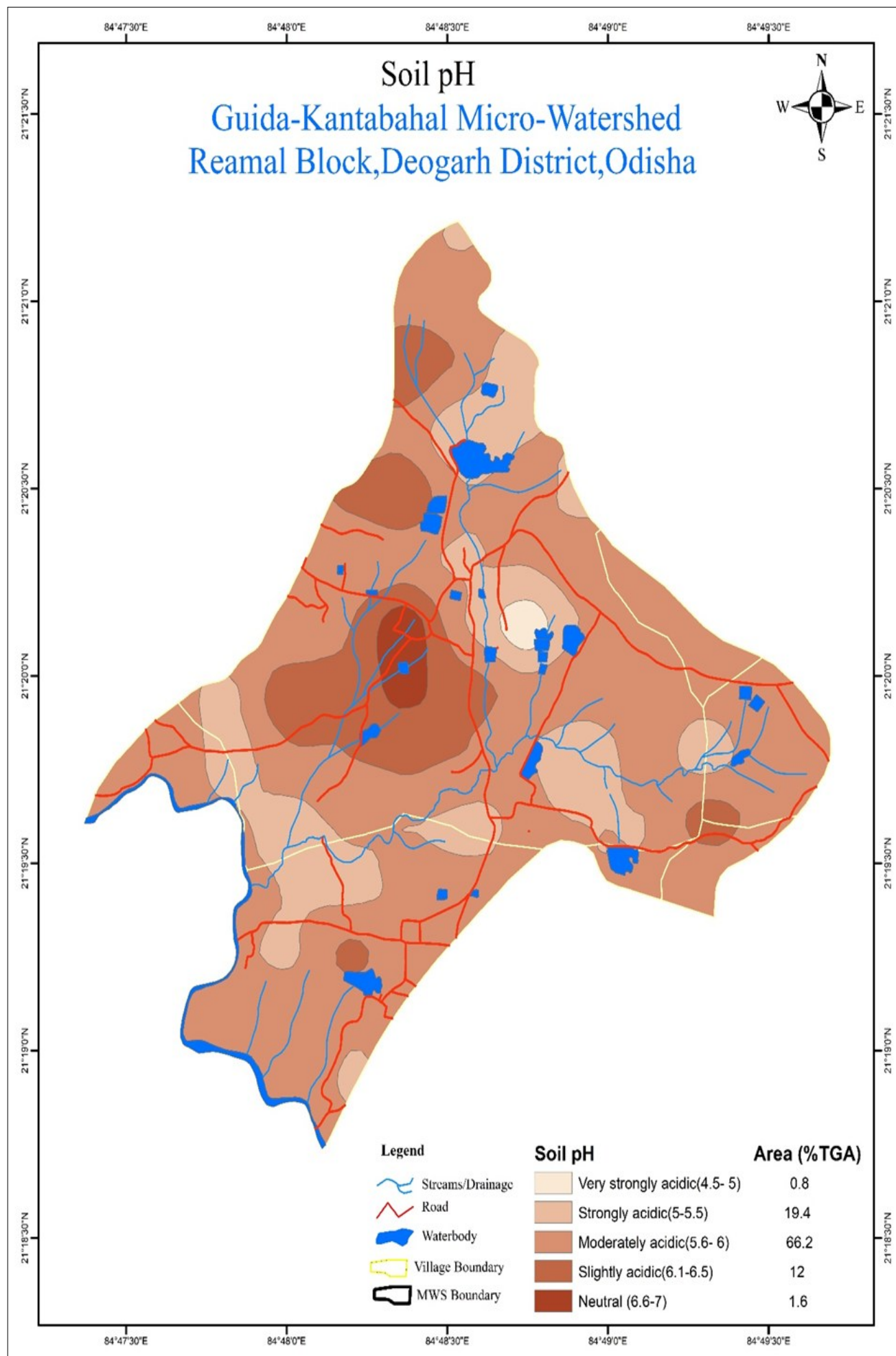


Fig. 7. Spatial distribution of soil pH across the Guida-Katabahal micro-watershed.

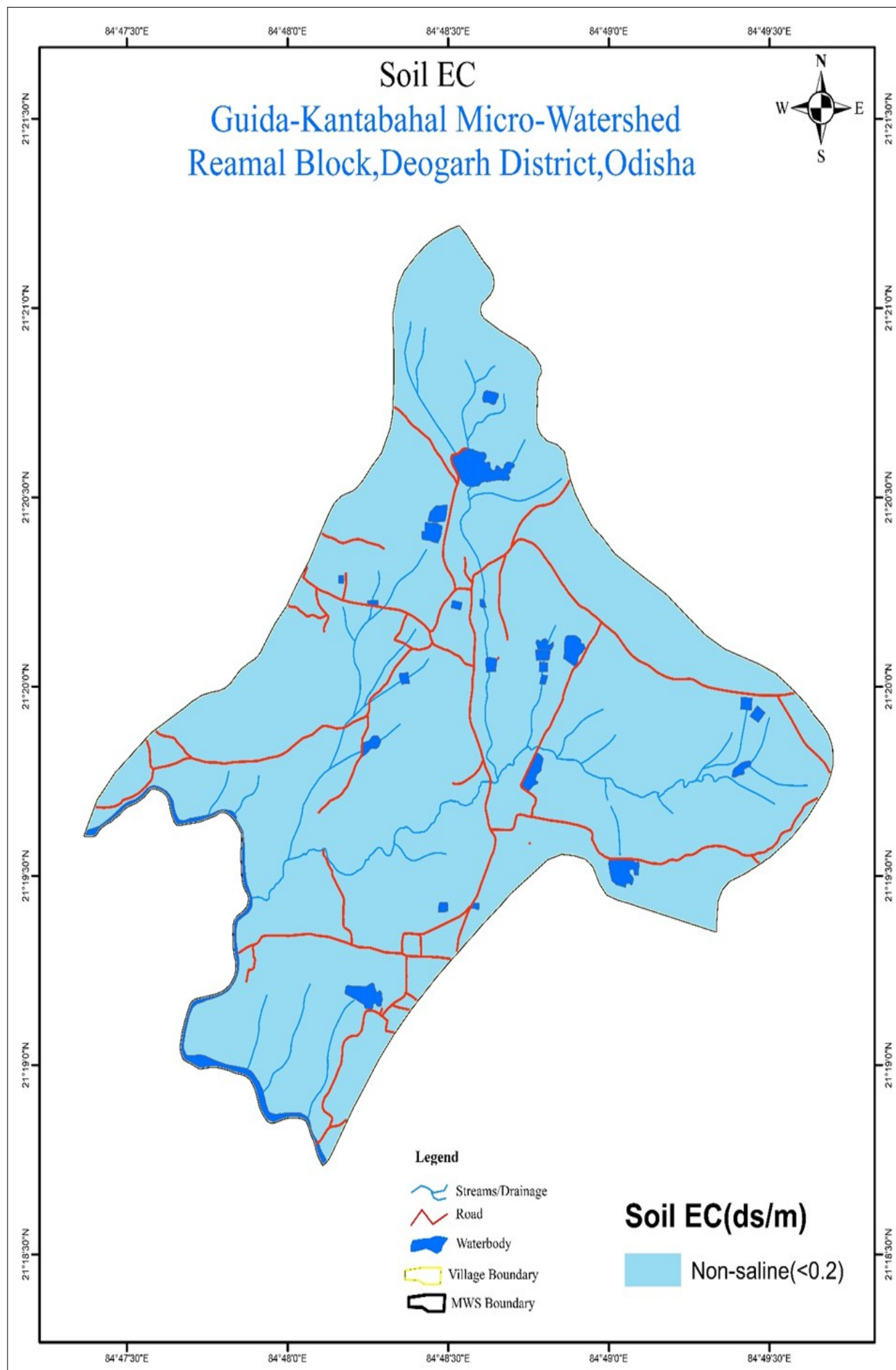


Fig. 8. Spatial distribution of electrical conductivity (EC) across the Guida-Kantabahal micro-watershed.

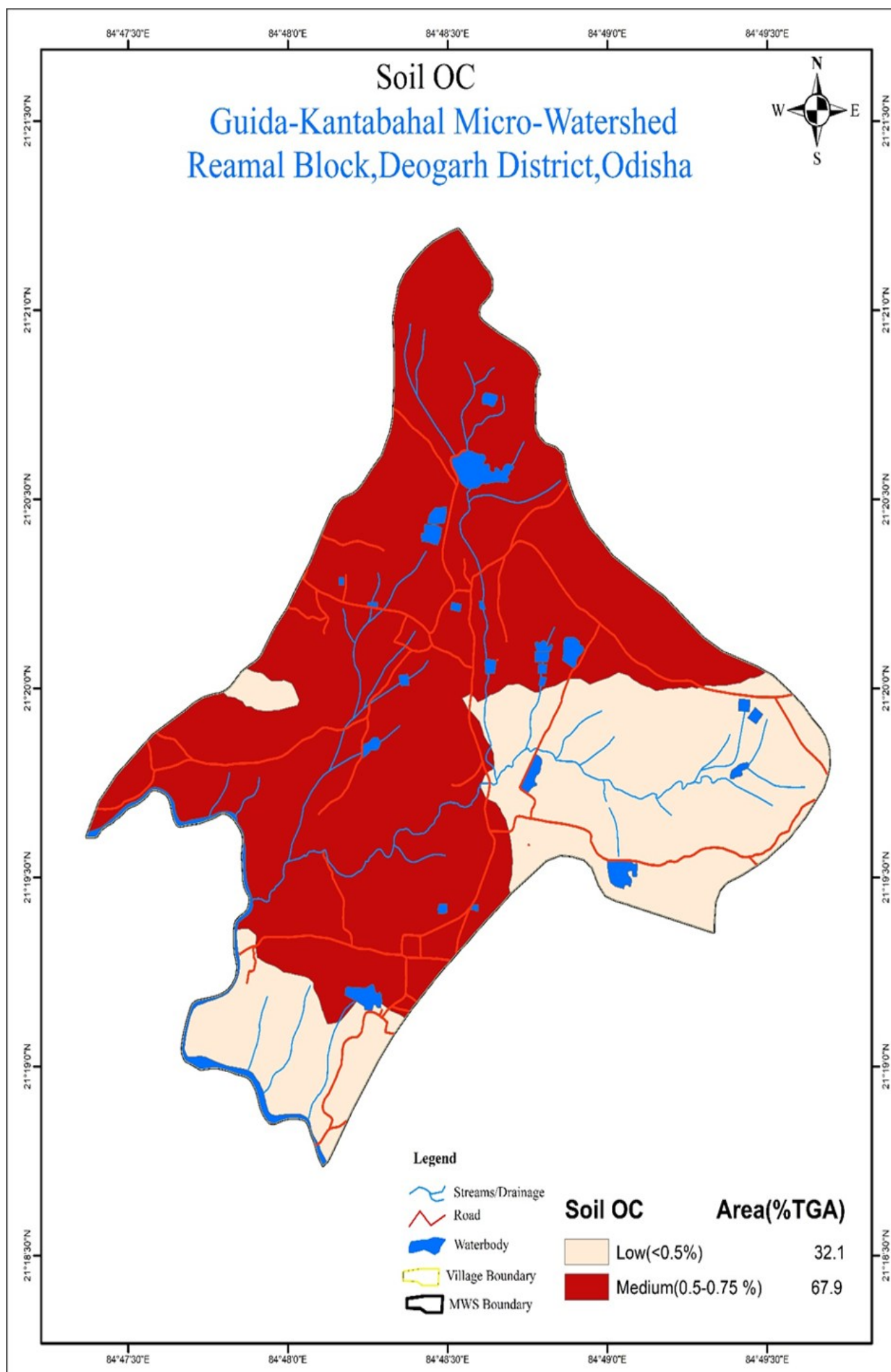


Fig. 9. Spatial distribution of soil organic carbon across the Guida-Kantabahal micro-watershed.

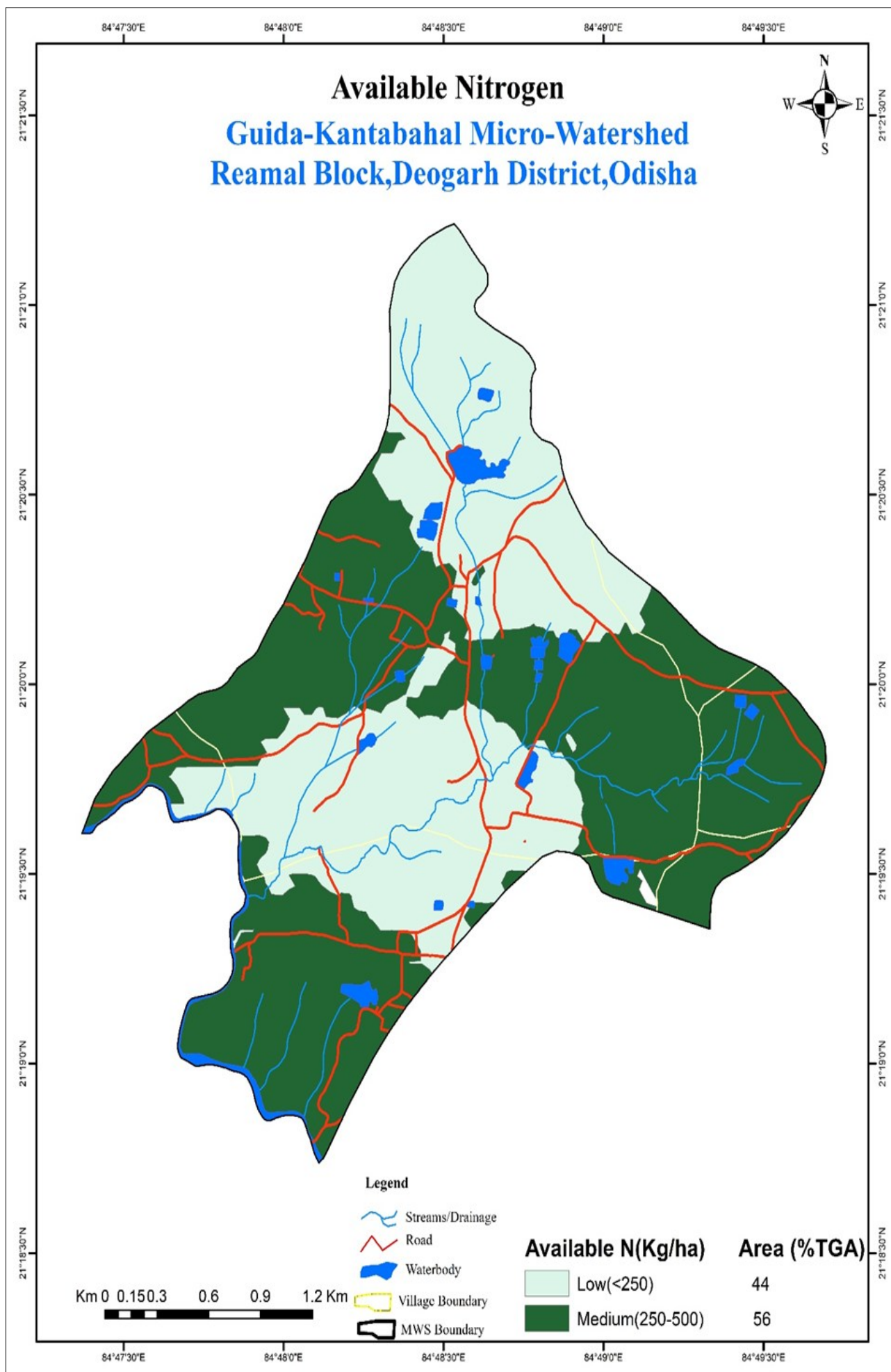


Fig. 10. Spatial distribution of available nitrogen across the Guida-Kantabahal micro-watershed.

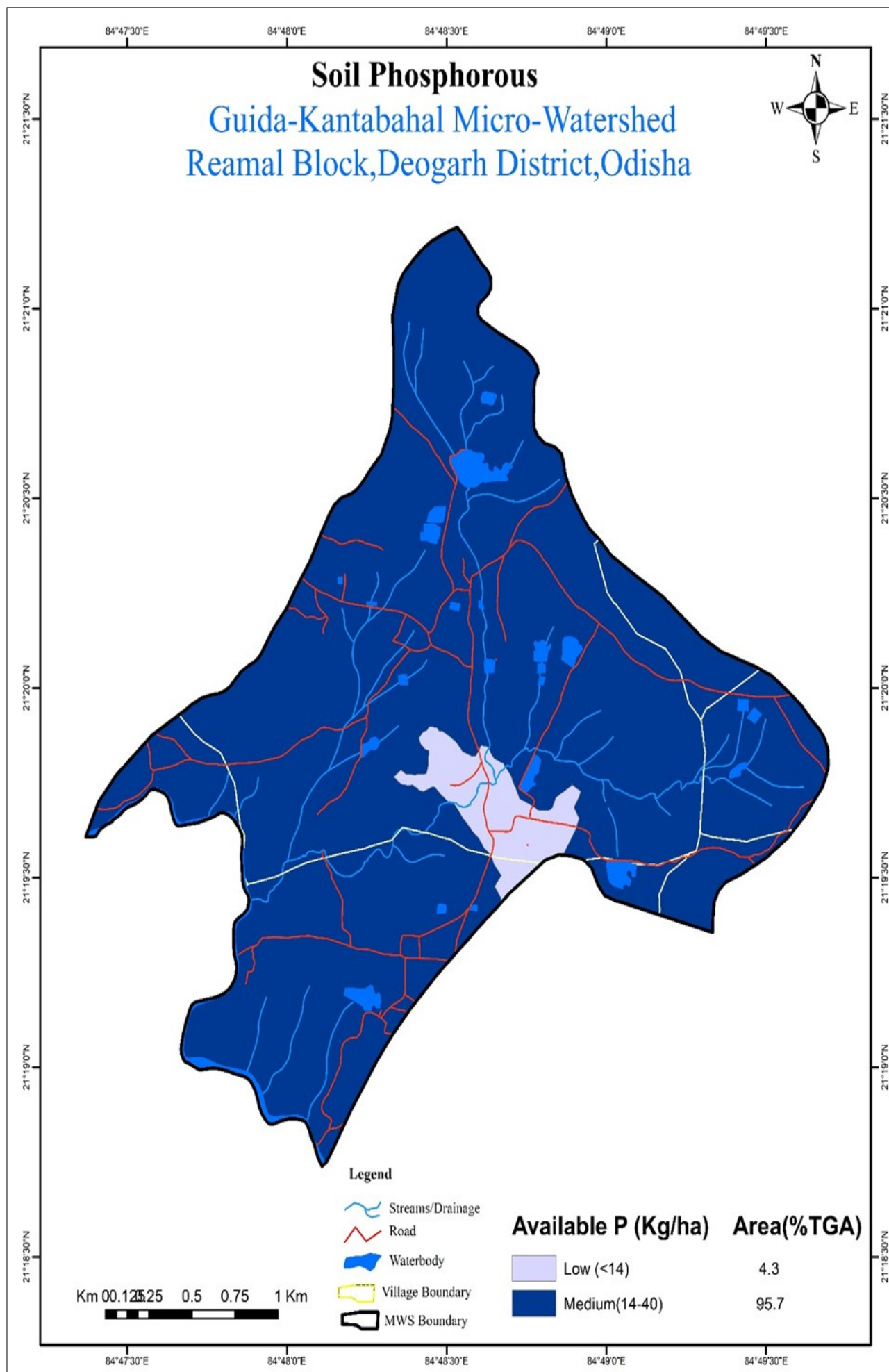


Fig. 11. Spatial distribution of available phosphorus across the Guida-Kantabahal micro-watershed.

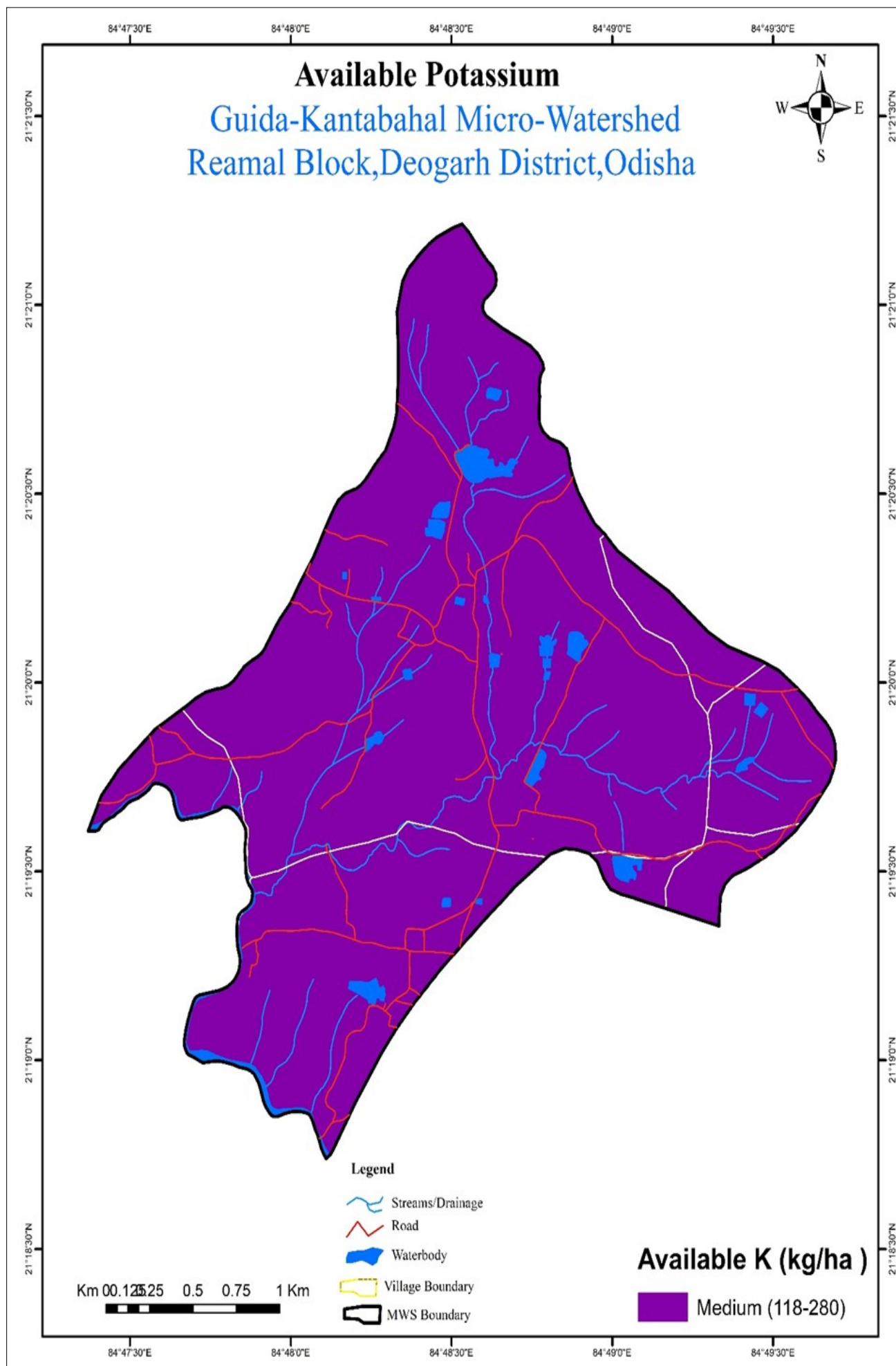


Fig. 12. Spatial distribution of available potassium across the Guida-Kantabahal micro-watershed.

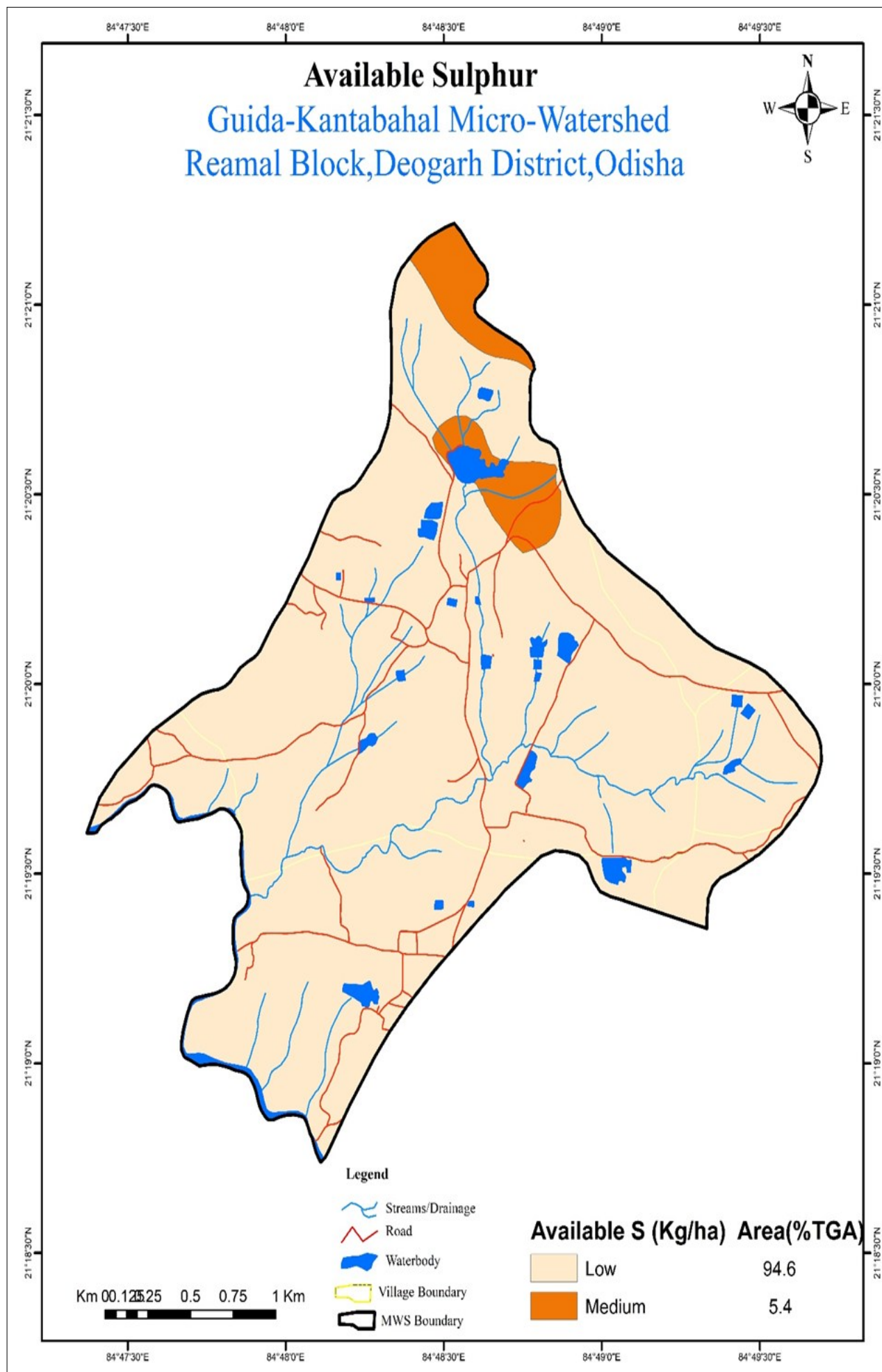


Fig. 13. Spatial distribution of available sulphur across the Guida-Kantabahal micro-watershed.

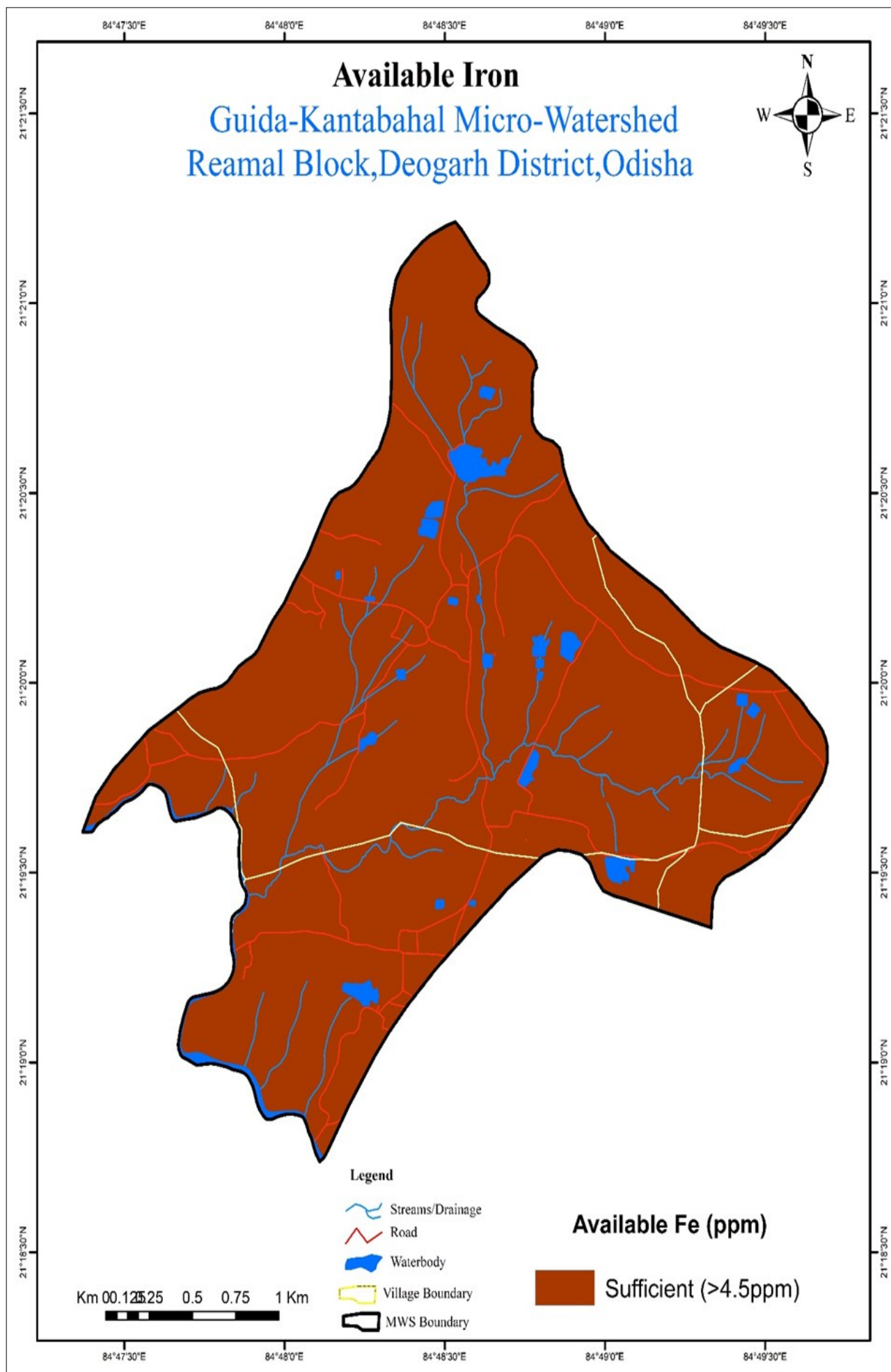


Fig. 14. Spatial distribution of available iron across the Guida-Kantabahal micro-watershed.

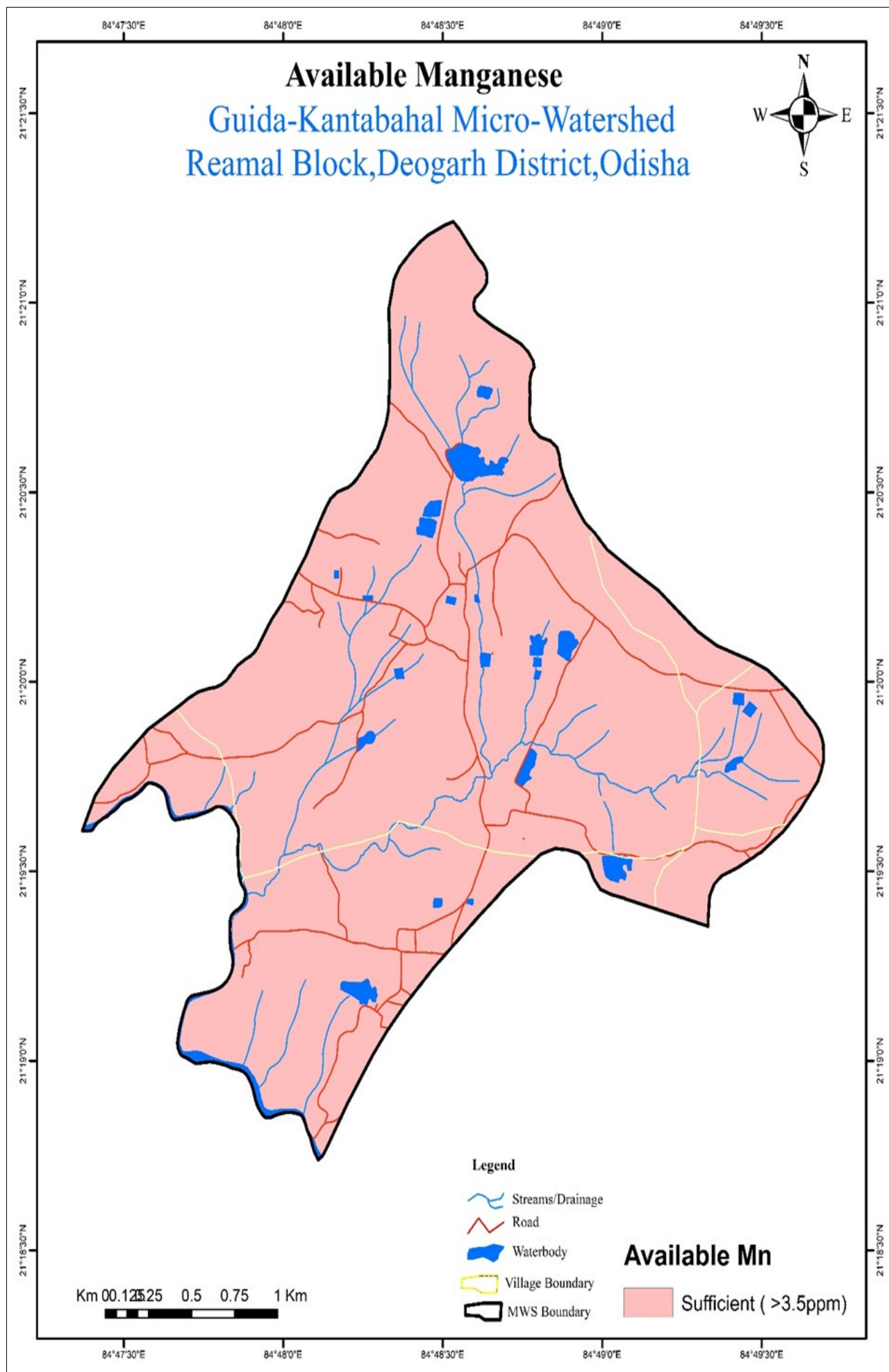


Fig. 15. Spatial distribution of available manganese across the Guida-Kantabahal micro-watershed.

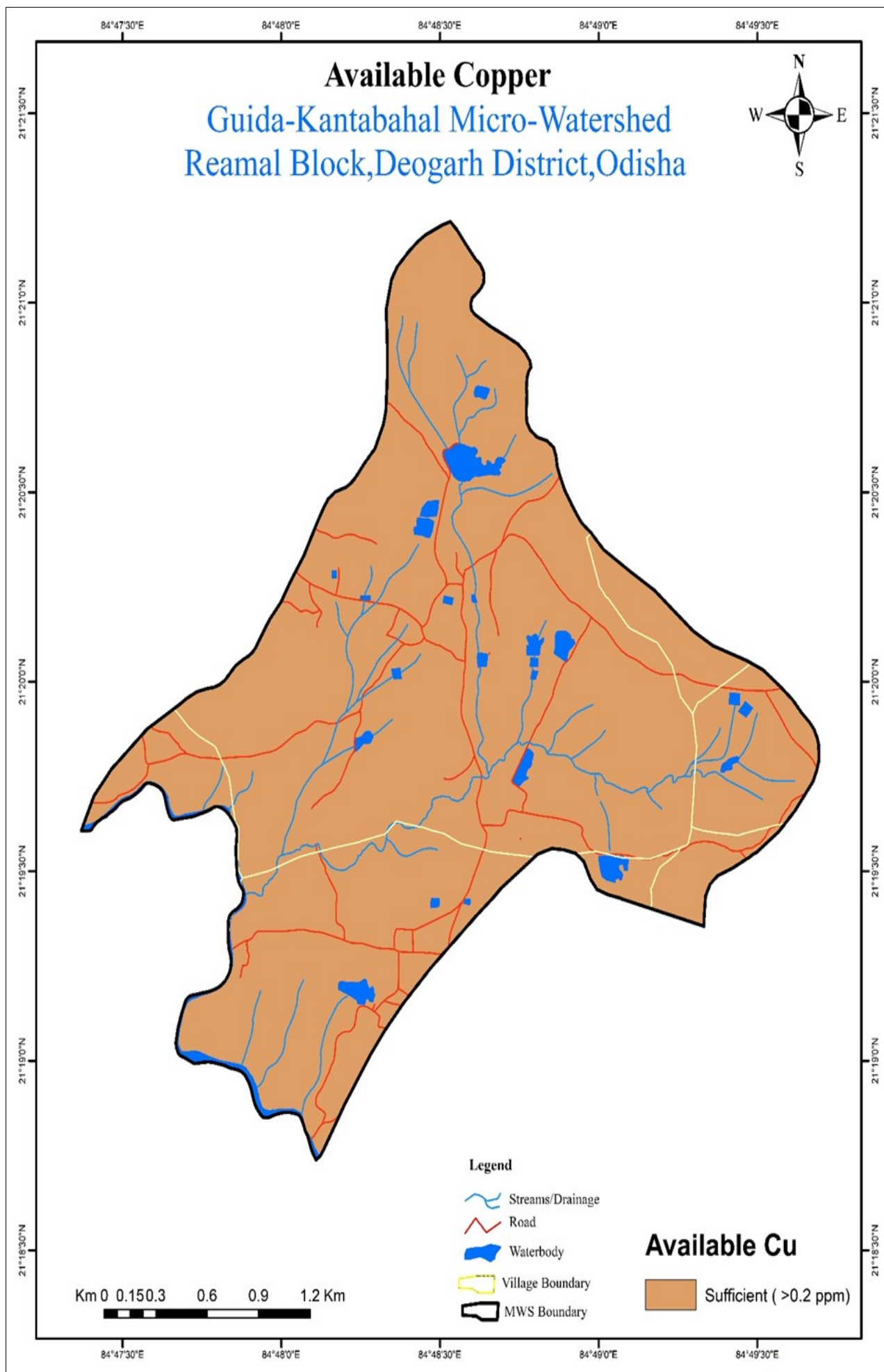


Fig. 16. Spatial distribution of available copper across the Guida-Kantabahal micro-watershed.

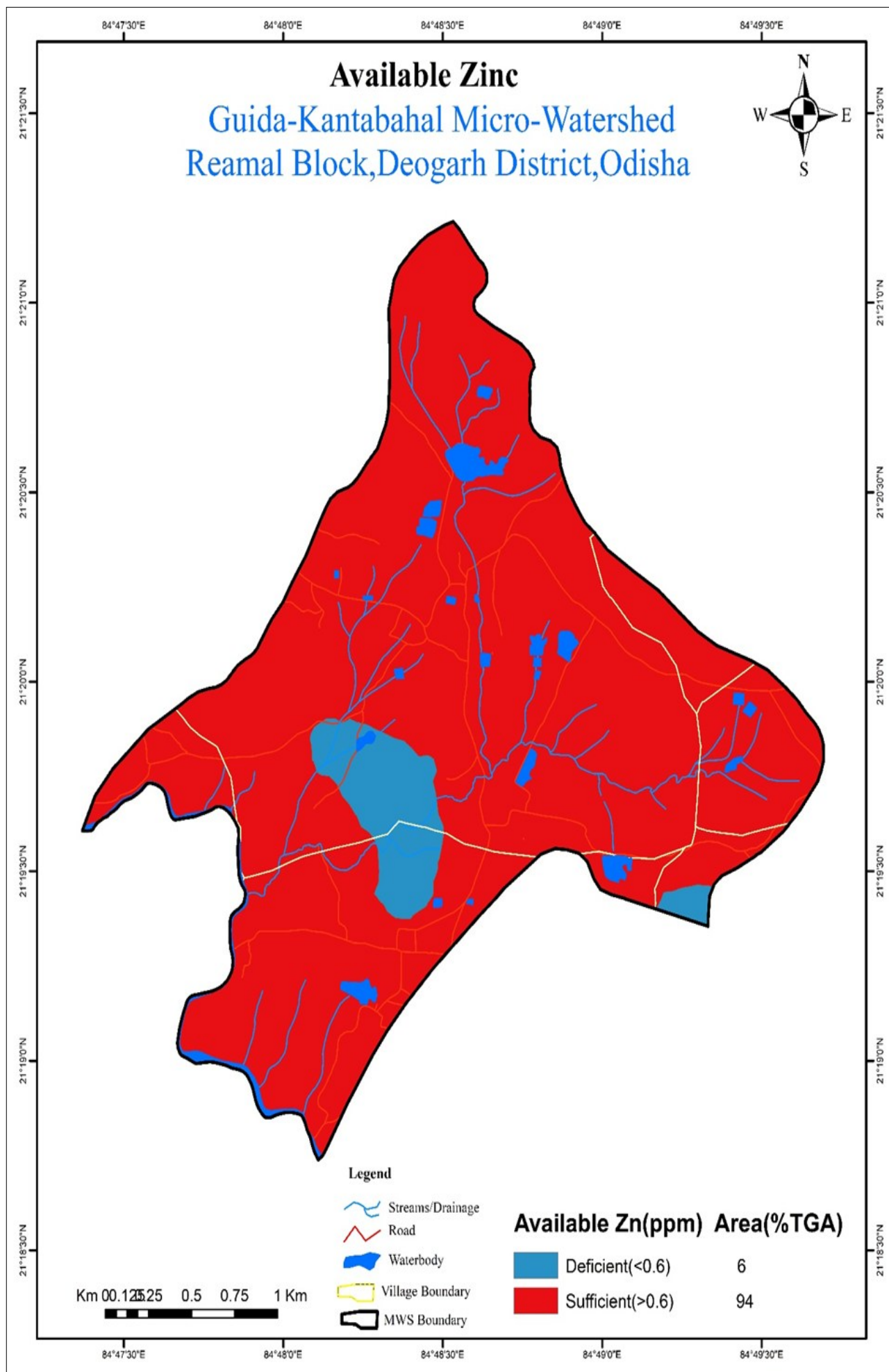


Fig. 17. Spatial distribution of available zinc across the Guida-Kantabahal micro-watershed.

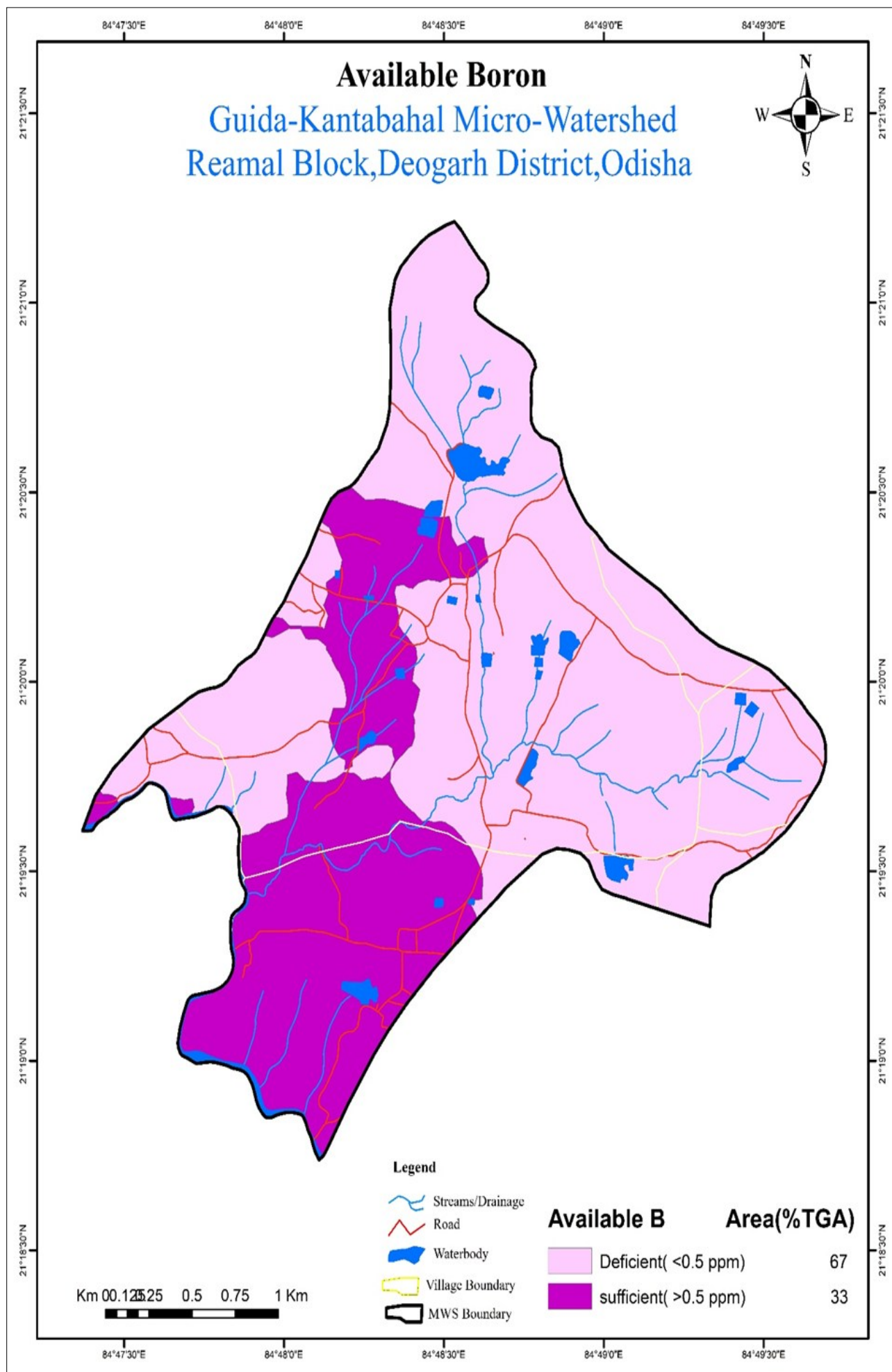


Fig. 18. Spatial distribution of available boron across the Guida-Kantabahal micro-watershed.

of management zones and provided a spatial framework for sustainable land resource management based on soil quality, moisture status, fertility attributes and erosion risk. The resulting spatial database and thematic maps are a powerful decision support tool for precision agriculture, watershed management and sustainable land-use planning. Machine learning, remote sensing and high-resolution spatial datasets could be integrated in the future to improve predictive assessment and to inform the process of intensifying agriculture in rain-fed agroecosystems while keeping climate change risks to a minimum.

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Authors' contributions

BRS and RN conducted the experiment, performed the experimental investigation and drafted the original manuscript. RN and SKR conceptualised the research idea and provided overall direction and supervision throughout the study. BSB and SS contributed to field investigation and data collection. PKS and RKN provided suggestions during manuscript revision. AD performed the formal data analysis while MKT contributed to critical revision and correction of the manuscript. All authors have read and approved the final version of the manuscript.

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During the preparation of this work the authors used Chat-GPT in order to improve language and readability, with caution. After using this tool, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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